Absence of Oospores of Downy Mildew of Grape Caused by *Plasmopara viticola* as the Source of Primary Inoculum in Most Western Australian Vineyards

B. X. Killigrew K. Sivasithamparam E. S. Scott

Plant Disease 89(7): 777 (2005)

DOI: 10.1094/PD-89-0777C

Grapevine downy mildew, caused by the obligate, oomycete pathogen, *Plasmopara viticola*, was first recorded in Western Australia (W.A.) in 1998 (2) and has subsequently been observed in most viticultural regions of the state. Heterothallism in *P. viticola* was established by Wong et al. (3), whereby more than one mating type of the pathogen is required for sexual reproduction to occur. Oospores are considered to be the source of primary inoculum for this disease with further, secondary infection being advanced by asexual inoculum. However, recent research in European vineyards suggests that the majority of infection throughout the growing season arises via sexually derived (oosporic) inoculum (1). Since downy mildew is relatively new to W.A., few surveys have been conducted to study populations of the pathogen within the state. It is also noteworthy that the incidence of oospores in Australian vineyards has not been reported. The objective of this research was to assess the occurrence and type of inoculum of *P. viticola* in W.A. vineyards. A total of 1,266 *P. viticola*-infected leaf discs (LD) from eight wine-grape (775 LD), five table-grape (450 LD), and seven unknown (41 LD) cultivars grown in 16 vineyards in 10 geographically separate regions of W.A. were collected in the growing seasons of 2001-2003. These regions range from Chittering in the north to Albany in the south and received 700 to 1,200 mm annual rainfall, mostly in winter. Each LD was cleared in 1 M KOH at 60 degrees C for 12 to 24 h and then was assessed for the presence of oospores with light microscopy. Leaves showing "mosaic"-type lesions (older infection) late in the season were collected where possible to ensure colony maturity and an increased likelihood of oospore formation. All LD from all regions were negative for the presence of oospores except for samples from a single vineyard (approximately 1,200 mm annual rainfall), where all 140 LD from six wine-grape cultivars contained oospores. The discovery that oospores were present in only one of 16 sampled vineyards provides a rare opportunity to study gene flow in field populations of the pathogen with time and to determine sources of primary inoculum where overwintering of *P. viticola* may not involve oospores.

References: (1) S. McKirdy et al. *Plant Dis.* 83:301, 1999. (2) A. Rumbou et al. *Eur. J. Plant Pathol.* 110:379, 2004. (3) F. P. Wong et al. *Plant Pathol.* 50:427, 2001.

Full-text record retrieved from the American Phytopathological Society article page on 12 September 2026. This is a locally formatted reading copy, not the publisher-typeset PDF.